

A Prefunded Design for the MultiPlanetary Exchange System

Design paper · Milestone 2 · Design v3

Abstract

We present a design for trading shares and a capped futures contract among the nine settlements of the MultiPlanetary Exchange System. Each settlement runs one exchange. A client who wants to trade at another settlement first transfers money to an account at that settlement's exchange over the backbone network. Once the money has arrived, the exchange executes the client's orders using only balances that it already holds. No exchange ever acts on value that is still in transit, so lost, delayed or duplicated packets can slow trading down but cannot create or destroy money or shares.

We evaluated the design with a discrete-event simulator of the network described in the brief. Without packet loss, an Earth client buys 1,000 Ares shares from a Mars seller and has them at home after 3.07 hours. With random loss, the median time is 3.07 hours, the 99th percentile is 7.70 hours, and all 2,000 simulated runs complete. The 300-hour capped futures contract settles correctly for both the rising and the falling price path, and both payouts are spendable at home after 328.17 hours. Clients at all nine settlements can use every product. Neptune is the slowest settlement: a share purchase from Neptune completes within 72 hours with probability 0.47 and within 168 hours with probability 0.97.

1 Introduction

The brief asks for a market system for clients on nine settlements, from Mercury to Neptune. Its scope covers equities, futures, options, bonds, currencies, lending, short positions, clearing and settlement. We implement and test equities, a capped futures contract, and the clearing and settlement of both; Section 2.3 explains how the other products fit the same framework and marks them as extensions. All communication between settlements is by radio. Messages take minutes to hours to arrive, some are lost, and some paths are blocked for weeks when they pass too close to the Sun. The main difficulty is that no exchange can know the current state of another exchange.

Our design avoids the need for that knowledge. Every account balance is held by exactly one exchange, and only that exchange can change it. An exchange only executes an instruction if the assets it needs are already in its own ledger. Money moves between exchanges as explicit transfers that are credited exactly once. As a result, the correctness of the ledgers never depends on whether a message arrives or when it arrives. Only the speed of service does. Unlike a distributed commit protocol [4], no exchange ever waits for another exchange to agree before it acts.

Section 2 describes the institutions and accounts. Section 3 gives the operating rules. Section 4 states the guarantees, the conditions they rely on and their cost. Section 5 lists the weaknesses we found by testing the design. Section 6 covers long-term operation, and Section 7 discusses what a real deployment would change. The evidence appendix contains the traces, calculations and simulation results behind every number in this paper. All results use the epoch 2026-09-22 00:00 TDB unless stated otherwise.

2 Architecture

2.1 Institutions

The system has nine institutions: one exchange at each settlement. The brief allows up to twelve, but we found no use for more. There is no clearinghouse. Each exchange keeps the accounts of its customers in its own ledger and is the only institution that can change those balances.

The three stock markets are located where the share registries and the initial sellers are: Ares Habitat on Mars, Belt Works on Ceres and Terra Fabrication on Earth. The capped futures contract is hosted by the Mars exchange. Mars is the best-connected settlement in the inner system (Appendix S3), and the price source, Bob, lives there.

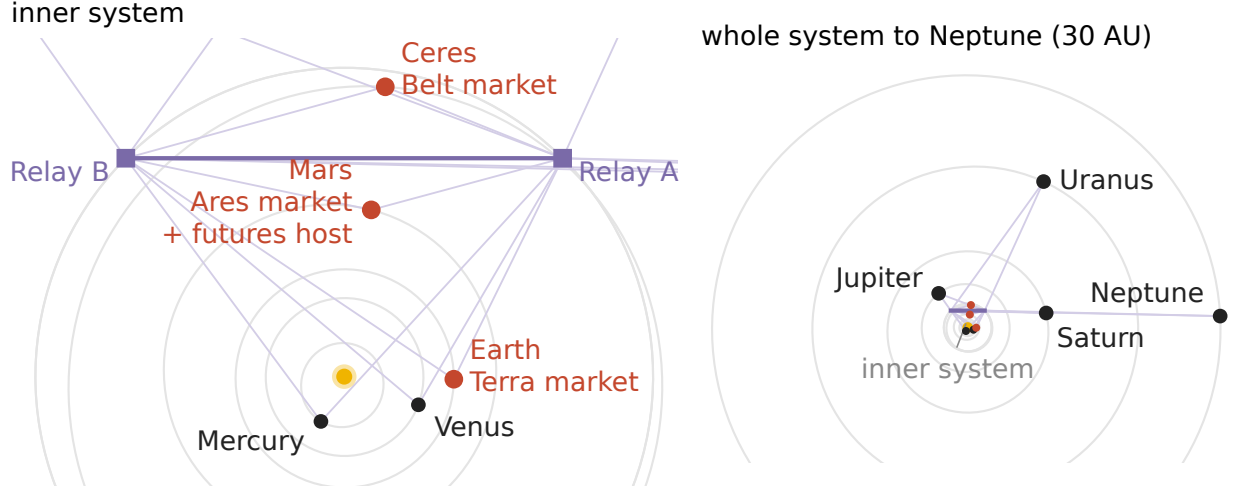


Figure 1: The nine settlements and two relays at the epoch. Purple lines are the 19 backbone links; red markers are settlements that host a market. Actual flights do not follow straight lines between the positions shown: each one is computed with the receiver moving during the flight, and a link is closed whenever its path passes within 0.1 AU of the Sun.

2.2 Accounts

Each client has one registered home settlement. A client may also hold a trading account at any of the other eight exchanges. Remote accounts start empty and receive assets only through explicit transfers. Opening a remote account does not create money, a new identity or any additional direct-service quota. Table 1 lists which institution may change each part of the shared state.

Table 1: Ownership of shared state. Each item has exactly one writer.

Shared state	Only institution allowed to change it
Available balances, order holds and contract escrow	The exchange that holds those balances
Order book and fills	The asset's market exchange
Debit and terms of an outgoing transfer	The sending exchange
Credit of an incoming transfer	The named receiving exchange, exactly once
Share registry (how many shares each exchange holds)	The asset's registry exchange, which is also its market
Contract positions, fixing and payout allocation	The contract host (Mars)

Every run starts from the balances in Table 2. The institutions themselves hold nothing.

Table 2: Opening balances at hour 0, all available at the home exchange. Remote accounts, holds, escrow, transit and the exchanges' own accounts are all zero.

Account	Alice	Bob	Cara	Dax	Eve	Fin	Total
Home	Earth	Mars	Ceres	Neptune	Uranus	Earth	5 settlements
NeoDollars	\$150,000	\$50,000	\$100,000	\$100,000	\$50,000	\$50,000	\$500,000
Ares	—	3,000	—	—	—	—	3,000
Belt	—	—	1,000	—	—	—	1,000
Terra	—	—	—	—	—	1,000	1,000

2.3 Products and scope

Table 3 lists every product in the brief's scope and marks each as implemented or as an extension. Every product, implemented or not, must follow the same two rules: an exchange acts only on assets already in its own ledger, and each balance has exactly one writer.

Table 3: Products in the brief’s scope. Extensions follow the same rules (prefunding, one writer per balance) but are not specified in full or tested.

Product	Status	Treatment in this framework
Equities (Ares, Belt, Terra)	Implemented, tested (S1a–c, S1i, S3)	Unit: one share, held at the registry or a custodian exchange. Limit orders at the market exchange (R2), fully prefunded. No end of life.
Capped futures	Implemented, tested (S1d–g, S2, E5)	Size Q per index point, strike 100, collar [60, 140], 300 h term, hosted at Mars, priced by Bob (R4, R5). Each side escrows its maximum loss, so the counterparty never bears a loss. Ends with the fixing and automatic payouts.
Clearing and settlement	Implemented	No central counterparty. Delivery versus payment in one ledger write at the host (R2); transfers between exchanges by R1.
Capped options	Extension	Writer escrows the maximum payout and the buyer prefunds the premium; settles like R4.
Short positions	Extension	The bounded futures short is implemented. Borrowed-share shorts need borrow, return and buy-in rules and are not supported.
Bonds and lending	Extension	A loan creates a claim and a liability, not cash. Guaranteed repayment needs escrowed backing; volatile collateral cannot guarantee it.
Currencies	Extension	A unit issued by one exchange and fully backed by escrowed NeoDollars, redeemable only there.

2.4 Example: an Earth client buys Mars shares

A cross-settlement trade proceeds in five steps. Figure 2 shows them for scenario S1a.

1. *Funding.* Alice instructs the Earth exchange, by local access, to move \$100,000 to her Mars account. Earth debits her balance immediately and sends a TRANSFER record over the backbone. While the record is in flight, the money still belongs to Alice, but nobody can spend it.
2. *Import.* Mars accepts the transfer exactly once, credits Alice’s Mars account and sends a RECEIVED record back to Earth.
3. *Order.* When Earth has the receipt, Alice knows that her money is at Mars. She then sends a signed order to Mars by the direct service. She sends several copies because direct packets are often lost.
4. *Trade.* Mars holds \$100,000 of Alice’s cash. It matches her order against Bob’s resting sell order and exchanges cash for shares in a single atomic step. Bob can spend the proceeds at Mars immediately.
5. *Delivery.* Alice asked for home delivery, so the same atomic step also creates a transfer of her 1,000 shares to Earth. For Alice the trade is complete when Earth credits the shares.

3 Operating rules

Each rule below states who acts, what information they may rely on and what happens when a message is lost or late. A change is *durable* if it survives a crash or reset. Every change of financial state described here is a single atomic, durable write to one exchange’s ledger.

3.1 R1: Moving assets between exchanges

A TRANSFER record carries a globally unique identifier (the source exchange and an export number), the owner, the asset, the amount, the destination and the purpose. Its terms never change after it is created.

1. The source exchange checks that the owner’s *available* balance is large enough. In the same write that creates the transfer, it debits that balance. From then on the value exists only as “in transit”.
2. The destination exchange credits the transfer the first time it sees its identifier and keeps a durable record of that identifier (compacted as described in Section 6). Any later copy only causes another RECEIVED.

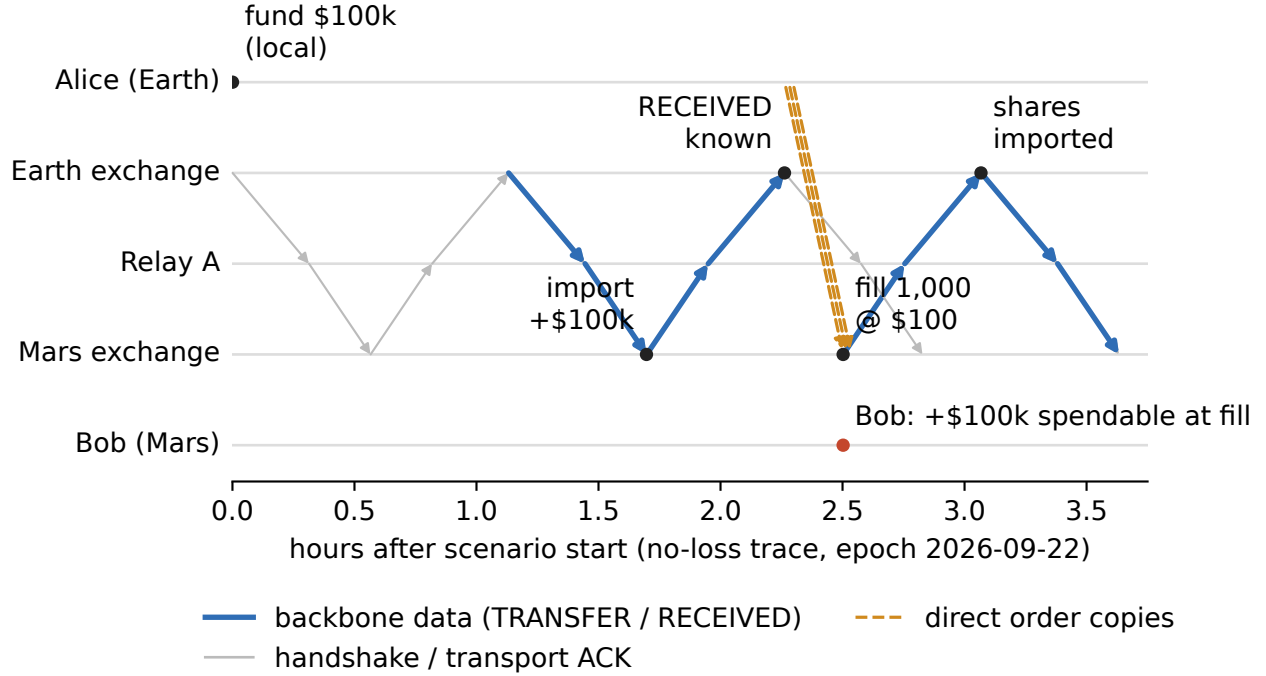


Figure 2: The S1a trade without packet loss. Bold arrows carry money or shares. Grey arrows are handshake and acknowledgement packets required by the network. Dashed arrows are Alice’s three direct copies of her order. The trade is complete at 3.067 h.

3. The source marks the transfer as delivered when the RECEIVED record arrives.

A missing receipt never causes a refund at the source. Instead, the courier keeps resending the same record: first through the network’s hop and endpoint retries, then by application resubmission after 24, 48, 96 and 168 hours, until the receipt arrives. If the destination account has been closed, the transfer is credited to a restricted balance and returned with a new transfer. Shares always move between the registry exchange and a custodian exchange, so the registry can reconcile how many shares each exchange holds.

3.2 R2: Orders, holds and matching

An order specifies its owner, a unique identifier, the asset, the side, an integer quantity, a limit price, a time-in-force (good-till-cancelled or immediate-or-cancel), an expiry no more than 30 days away, and a disposition. The disposition says whether results stay at the market or are sent home automatically.

The market accepts an order only if the owner’s available balance at that exchange covers it: quantity times limit price for a buy, or the shares for a sell. The covered assets are moved into a hold that nothing else can use. An order that is not covered is rejected permanently, and the client must fund the account and send a new order with a new identifier.

Matching uses price-time priority at the resting order’s price, and self-trades are skipped. Each fill is one atomic write. It reduces the buyer’s cash hold by the quantity times the limit price and the seller’s share hold by the quantity. It credits the buyer with the shares and with any price improvement, and credits the seller with the quantity times the trade price. If the buyer chose home delivery, the same write creates the transfer home. Cash and shares therefore change hands together, which is delivery versus payment within one ledger [3]. A cancel releases only the unfilled part of an order. If a cancel arrives before its order, the exchange records it so that the order can never open later. Orders cannot be amended; the client cancels and sends a new order instead.

3.3 R3: Withdrawals and account closure

A WITHDRAW instruction names an amount or ALL. ALL is evaluated once, when the host first processes the instruction, and repeated copies return the same transfer. A withdrawal can only take the available

balance. It never cancels orders and never touches margin. A CLOSE instruction stops new orders, cancels open ones and withdraws the free balance. Escrowed margin stays at the host until the contract pays out, and the payouts are then sent home automatically. The home exchange cannot release money held at another exchange; only the exchange that holds it can send it back.

3.4 R4: Capped futures and the margin rule

The long and the short agree on a size Q in NeoDollars per index point, a strike $K = 100$ and a collar of $[60, 140]$. The long's payoff over the life of the contract is

$$Q \times (\min(\max(P, 60), 140) - 100),$$

where P is the fixing price, and the short receives the negative of this amount.

Margin rule. From opening until fixing, each side must hold its maximum possible loss, $40Q$, in escrow at the host. There is no variation margin, no margin call, no netting and no share collateral. The rule is fixed before any price is observed. Because the escrow equals the largest amount either side can ever owe, no price path can produce an unfunded obligation. The rule is the same for the rising and the falling price path.

To enter the contract, each party funds its Mars account (R1) and then sends a contract instruction with the full terms. Mars moves that party's margin from its available cash into a hold. When both holds exist, the contract opens atomically and its maturity is set to 300 hours after opening. If both margins are not held by the opening deadline, 168 hours after the scenario starts, the contract is rejected. Each hold then returns to the party's available balance at Mars, and a client who has not received an OPEN report withdraws it. Payouts are transferred home in the same write as the fixing.

3.5 R5: Price source and fixing

Bob is the named index source at Mars. He learns the observation schedule locally when the contract opens and publishes one signed observation by local access at 60, 120, 180, 240 and 300 hours after opening. Mars fixes the contract 24 hours after maturity using observation 5. If observation 5 is missing, or if two conflicting copies exist, Mars uses the latest earlier valid observation, or 100 if there is none. Observations that arrive after the fixing are ignored. The 24-hour grace period allows a remote price source's packet to arrive before the fixing. With a local source it only costs one day of escrow (3.84 million dollar-hours) and could be removed (Section 5).

3.6 R6: Communication services

Table 4 shows which service carries each type of message. The nine exchanges divide the 600 backbone originations allowed per day into 66 each (594 in total), so no shared counter is needed. SYN packets, first transmissions, RECEIVED records and application resubmissions count against the quota. Automatic retries and ACKs do not, although our packet totals include every launch. Client instructions never use the backbone, and exchanges never use the direct service to coordinate with each other.

Quota classes. A data packet that carries any record other than RECEIVED is *routine* and may be sent only while the exchange has made fewer than 44 originations in the last 24 hours. SYN packets, receipt-only packets and resubmissions may use all 66, so at least 22 a day always remain for recovery, however much new work is queued.

Packet sizes. Every record is a fixed-width binary layout with an 88-byte common header (type, length, identifier, creation time and the issuer's Ed25519 signature). A TRANSFER is 176 bytes, a RECEIVED 120 bytes and a client instruction 128 to 256 bytes. A packet has the brief's 64-byte header and at most 960 bytes of payload, of which 16 are our sub-header, so it carries at most 944 bytes of whole records: five transfers, seven receipts or three contract instructions. Records are never split. The sender closes a packet when the next record would not fit, and a client batch that does not fit is sent as several direct packets, each counted in the 12. Appendix F gives every field.

Status reports. A remote client learns that its margin is held or that its contract is open from its own home exchange. When a contract changes state, Mars adds one entry per affected remote client to a STATUS record for that client's home exchange. All entries for the same home exchange created at the same moment travel in one record, so the cost grows with the number of home exchanges, not with the number of clients. The home exchange tells the client by local access. Status therefore uses Mars's backbone quota of 66

originations per day instead of its 12 direct packets, gets the backbone’s retries instead of a single direct copy, and still reaches a client whose own direct path is behind the Sun. The price is packets: each STATUS record crosses two backbone hops with their receipts and acknowledgements, which adds about 20 launches to a two-party contract (Appendix S1).

Table 4: Which communication service carries each message.

Message	Service	Quota charged
TRANSFER (funding, withdrawal, delivery, payout) and RECEIVED	Backbone, on a fixed route of at most 3 links	Sending exchange, 66 per rolling 24 h
Order, cancel, withdrawal, close, contract instruction	Local access at the host; otherwise direct	The client’s own 12 per 24 h, at least 60 s apart
Price observation	Local access at Mars (Bob is local)	None
Contract status for remote clients (HELD, OPEN, REJECTED)	Backbone STATUS record from Mars to the client’s home exchange, which tells the client by local access	Mars exchange, 66 per 24 h; one record per home exchange

3.7 R7: Client policy

Clients are programs that know the published ephemeris but not the hidden incidents. They follow four rules.

- *Fund first, then instruct.* A client sends an order or contract instruction only after its home exchange has recorded the RECEIVED for the funding transfer. The client learns this by local access, so the order cannot be rejected for lack of funds.
- *Copies.* The client sends the smallest number of direct copies, 60 seconds apart, that gives at least a 99% chance that one arrives, up to a maximum of 8. From Earth to Mars this is 3 copies, which are all lost with probability 0.0021. From Neptune to Mars it is 8 copies, which are all lost with probability 0.46.
- *Fallback.* If an equity buyer’s shares have not arrived home after twice the combined direct and backbone delay plus 6 hours, the client resends the order with the same identifier until the order expires. Duplicates are harmless. A futures party that has no HELD or OPEN status after twice the direct flight time, plus three backbone route times from the host to its home, plus 6 hours (at most 12 hours, so that a distant client uses its whole daily quota), resends its instruction until the deadline.
- *Access check.* Before funding, the client checks that it will be able to reach the host for the whole planned sequence, either directly or by forwarding (R9). If it will not, the client waits for the next window that is long enough. If that window starts after the deadline, the client does not trade, and its money stays at home.

3.8 R8: Recovery

Ledgers, outboxes and identifier records are durable. After a reset, an expired session or a delivery that the network gives up on, the courier opens a new session and resubmits the same records after 24, 48 and 96 hours and then every 168 hours. It waits out known closures and quota limits, and it gives recovery traffic priority over new traffic. Table 5 summarises how each party behaves when a message goes wrong.

3.9 R9: Forwarding through a third exchange

The Sun sometimes blocks a client’s direct path to the host for weeks, while the backbone stays connected. Client instructions cannot use the backbone, so R9 gives them a second direct route. If the client’s direct path to the host is closed when an instruction is due, the client sends it instead to the exchange of a third settlement. It picks the settlement whose two direct legs are both open and which gives the earliest arrival at the host.

Table 5: What each party knows and does when a message goes wrong.

Event	Sender	Receiver	Financial effect
TRANSFER lost or late	Has no RECEIVED yet and keeps resending the same transfer	Knows nothing; the money cannot be spent there	The value is in transit, still owned by the client
RECEIVED lost	Resends the TRANSFER	Ignores the duplicate and resends RECEIVED	None; the money is already spendable at the host
Duplicate transfer, order or instruction	—	Returns the original result; a copy with different terms is rejected	None
All order copies lost	Resends the same order ID when its fallback timer expires	No order exists; the cash stays available	The client’s cash is idle at the host
Cancel lost or late	Assumes nothing until it sees the order state	The order may still fill	A fill stands; the cancel releases only the unfilled part
Conflicting price copies	—	Treats the observation as invalid and uses an earlier one	The fixing is final and never revised
STATUS record lost	Courier resends it like any record	Client fallback resends the instruction; the duplicate only triggers a fresh status	None
Host isolated for 72 h	Remote clients cannot send instructions	Local trading, matching and maturity continue	Payout transfers wait, already funded, in the outbox
Exchange reset	Loses its open sessions	Ledger is intact; the outbox is resent after a backoff	None
Forwarding exchange slow or silent (R9)	Resends the instruction when its fallback timer expires	Host sees nothing, or a later duplicate it ignores	None; the client’s funds stay available at the host

1. The client wraps its signed instruction in a FWD record with a forwarding attempt number $k = 1, 2, \dots$ and sends it to the third exchange by direct service, with the number of copies set by R7 for that leg.
2. The third exchange forwards each new pair (instruction identifier, k) once, by direct service under its own quota of 12 packets per day and with the number of copies set by R7 for the second leg. Further copies of the same attempt are ignored. It keeps only the pair, until the instruction’s expiry or opening deadline.
3. The host treats the forwarded instruction exactly like a direct one. It checks the client’s signature, balance and instruction identifier, so a forwarded copy and a direct copy of the same instruction are duplicates.

The forwarding exchange gains no authority. It cannot change a signed instruction, no asset passes through it and it owes nothing financially. The client learns the result from its home exchange over the backbone (R6), which works during the closure. If no status arrives before the R7 fallback timer, the client sends attempt $k + 1$, through the third exchange that is best at that moment (possibly a different one) or directly once the path reopens. A forwarded instruction uses two direct legs, so it costs about twice as many packets, and these are counted in every total. During the 2028 Mars conjunction, Earth clients cannot reach Mars directly for 53 days. With R9, the S1a trade completes in 4.55 hours instead of not taking place (Appendix E5).

4 Guarantees and costs

We distinguish between properties that the *rules* guarantee in every possible run and properties that the *model* (orbits, loss law and quotas) makes likely. The first kind holds for any pattern of packet loss, delay or reordering. The second kind is supported by simulation evidence but is not proven. Table 6 lists both.

Table 6: Guarantees of the design and the conditions they depend on.

	Guarantee	Reason	Condition
G1	The total of NeoDollars and of each share type never changes	Every write debits and credits equal amounts; value in transit is a ledger state	Rules only; checked after every event of every traced run
G2	No asset is used for two purposes at once	Available, held, escrowed and in-transit balances are separate	Rules only
G3	Every futures payout is fully funded	Escrow equals the maximum loss and is posted before opening	Rules only; holds for any price path
G4	Each transfer is credited exactly once	The destination keeps a durable record of every credited identifier (tombstone or watermark)	Rules and durable storage
G5	A fill is final and immediately spendable at the host	Both holds exist before the atomic fill	Rules only
G6	A funded transfer eventually arrives	Retries never stop and the backbone is always connected (E4)	Model; under random loss there is no fixed deadline
G7	Completion times	Measured in simulation	Model; see Table 7 and Appendix E2 and S3

4.1 Service at every settlement

Every product is available at every settlement. Clients can fund and withdraw at all three markets, trade Ares, Belt and Terra shares, and enter the capped futures at Mars. What differs between settlements is the speed of service and the probability of completing within a given time. Table 7 shows these figures for each settlement.

Table 7 uses the Ares purchase. Appendix S3 repeats it for every product at every settlement. A purchase is local, with no network traffic, only where the client’s home hosts that market: Ares and the futures at Mars, Belt at Ceres and Terra at Earth. Everywhere else the three share purchases complete within 1.1 hours of each other without loss, because the time is set by where the client is, not by the product.

Every futures party must reach Mars before the 168-hour opening deadline. Parties in the inner system almost always do: in all 2,000 simulated runs with random loss, the contract opened (Appendix E2). For a Neptune party (Dax long against Cara), the contract opened before the deadline in 99.7% of runs with random loss. In the other runs it was rejected at the deadline and all margin was returned (Appendix S3).

4.2 Costs

A single cross-settlement trade with home delivery uses 47 packets: 44 backbone launches, including handshakes, receipts and ACKs, and 3 direct copies. It uses 5 backbone originations from the quotas. Repeated trading is much cheaper per trade. In the \$10,000 example (S1i), one funding transfer supports four fills, a cancellation, a race between a cancel and a fill, and a final transfer home, using 59 packets in total. Table 8 gives both efficiency measures with one definition of a transaction (Appendix S1).

The brief’s measures count every leg, so one \$100,000 purchase settles \$300,000. For the client, S1a costs 47 packets and commits exactly the price paid (ratio 1.0); the two futures sides post 80% of the notional (40% each). Remote funding moves the client’s money to its own account; the purchase completes only when the shares are spendable at home, so the component ratio does not mean a third of the price is needed (Appendix S1).

The futures contract locks \$160,000 for 324 hours. This is 51.84 million dollar-hours of escrow, and at its peak 80% of the notional value is encumbered. An equity purchase ties up its full value while it is in transit

Table 7: Service at each settlement at hour 0. Backbone min: one-way delay in minutes to the Mars, Ceres and Earth hosts. Copies, P: direct copies per instruction to Mars and the probability that at least one arrives. The next three columns are for a \$100,000 Ares purchase with home delivery: completion time in hours without loss, the median under random loss, and the probability of completing within 72 hours (1,000 runs each; Appendix S3).

From	Backbone min	Copies, P	No loss	Median	P(72 h)	Notes
Mercury	42 / 42 / 46	3, 0.997	3.8	3.8	1.00	Direct path to Mars closed from h 22 to h 357; instructions are forwarded (R9)
Venus	37 / 36 / 41	3, 0.997	3.3	3.3	1.00	Closed near Venus–Mars conjunctions; instructions are forwarded (R9)
Earth	34 / 33 / local	3, 0.998	3.1	3.1	1.00	Terra local. Direct path to Mars closed about 53 days every 2.1 years; forwarded (R9)
Mars	local / 30 / 34	local	0.0	0.0	1.00	Ares and futures local; Belt (Ceres) and Terra (Earth) remote
Ceres	30 / local / 33	2, 0.992	2.6	2.6	1.00	Belt local; Ares and futures (Mars) and Terra (Earth) remote
Jupiter	40 / 41 / 51	4, 0.992	3.9	4.0	1.00	4 copies per instruction
Saturn	76 / 75 / 79	7, 0.990	7.6	10.6	1.00	7 copies per instruction
Uranus	155 / 154 / 158	8, 0.885	15.4	26.8	0.99	8 copies; the fallback is often used
Neptune	247 / 246 / 250	8, 0.545	24.7	74.4	0.47	8 copies still all fail 46% of the time

Table 8: Efficiency as defined in Section 7 of the brief (no loss). A transaction is a completed transfer, trade fill or settled contract; exposure is each side’s largest possible loss to its counterparty.

Scenario	Tx	Value settled	Packets	Pkts/tx	Peak enc.	Cap. eff.	Exposure
S1a Equity, Earth buys Ares	3	\$300,000	47	15.7	\$100,000	0.33	\$0
S1c Equity, Uranus buys Terra	3	\$120,000	52	17.3	\$40,000	0.33	\$0
S1d Futures, either path	5	\$520,000	141	28.2	\$160,000	0.31	\$80,000
S1f Futures, Q = 2,500	5	\$650,000	141	28.2	\$200,000	0.31	\$100,000
S1i Repeat trading	7	\$23,020	59	8.4	\$10,000	0.43	\$0

or on order, and nothing after the trade. No institution needs to hold capital. Figure 3 shows where Alice’s and Cara’s cash is during the contract.

5 Risks

Table 9 lists the weaknesses we found, ranked by their expected harm under our own evidence. The most important ones are interactions between our rules and the orbital geometry. We did not find them by inspecting the design. They came out of the incident search, the runs at shifted epochs and the runs from the outer settlements. Changes I1 and I2 are already part of the design evaluated in this paper.

6 Long-term operation

The backbone never disconnects. The two relays are 90° apart on the same circular orbit, and the link between them always passes at least 2 AU from the Sun. A link between a settlement and a relay is blocked only when the settlement is almost directly behind the Sun as seen from that relay, at an angle of at least 159°. This cannot happen for both relays at the same time. Every settlement therefore always has a route of at most three links to every other settlement, except during maintenance and incidents. Our 200-year scan confirms this: every settlement had a route to every market host at all 1,753,201 hourly samples. The slowest one-way route observed was Mars to Neptune at 315 minutes. Because a route has at most three links, its

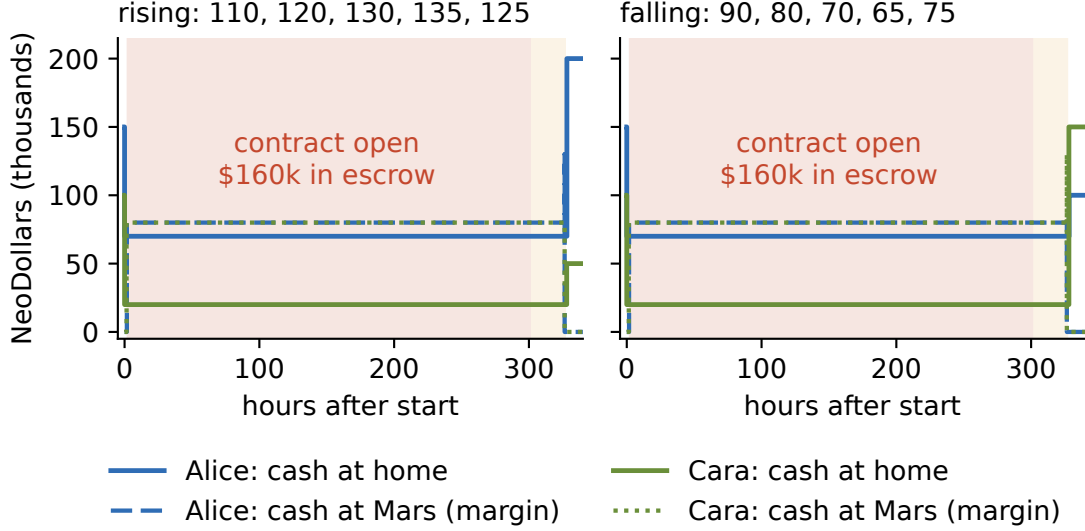


Figure 3: Location of Alice’s and Cara’s cash during the futures contract, without packet loss. The red band marks the period when the contract is open and the beige band the fixing grace period. Both price paths use the same margin and end with \$500,000 in total.

length is at most $(r_a + R) + 4 + (r_b + R)$ AU at any time, where r_a and r_b are the aphelion distances of the two settlements and $R = 2.83$ AU is the relay radius. With empty queues this bounds every one-way route delay for all time; the largest bound is 358 minutes (Neptune to Ceres). Direct paths, in contrast, do close for weeks at a time during conjunctions, so every promise that depends on the direct service is conditional.

Long outages and the backlog afterwards. Only two things have a deadline: orders, which expire after at most 30 days and release only their own hold, and contract opening deadlines, which return the margin. Transfers, receipts and payouts wait in durable outboxes. Sessions that are idle for 7 days are reopened. A packet that reaches the network’s 30-day lifetime is replaced by an application resubmission with the same financial identifier. When a route reopens, each exchange works through its outbox in packets of five transfers, within its 44 routine originations per day. We say a backlog has *cleared* when every transfer is credited at its destination and the source holds every RECEIVED. In simulation (Appendix E6), with no competing traffic, 300 transfers from Earth to Mars are credited within 25 hours and cleared within 25 hours of the route reopening, and in at most 30 hours in 20 runs with random loss. Other work from the same exchange shares the 44 routine originations: 300 transfers to each of the eight peers clear in about 10 days.

Obligations further in the future. A contract stores its maturity and fixing times as absolute times. The fixing is a local computation at the host, and the payout transfer is created in the same write. A maturity several years away therefore needs no message in order to take place.

Capital. There is no endowment. New capital can enter only through fees or contributions that are debited from real accounts and moved by R1. Escrow is released only by a fixing or a rejection, so capital cannot leak out of the system or be created.

Storage and identifiers. An identifier is 12 bytes: a 32-bit principal prefix and a 64-bit counter. At a million identifiers per second one principal’s counter lasts about 585,000 years. Times are stored as signed 64-bit integer milliseconds, which covers about 292 million years in each direction. Our simulator uses floating-point seconds, which can no longer resolve 1 ms after 142,710 years, so a production system must use integer time and extended-precision orbital phase. State is bounded: at most 32 live orders per client per host, 256 per host, and 4,096 export slots per host, each an unresolved transfer or a reservation. A remote client’s contract instruction reserves its payout’s slot and a home-delivery order its delivery’s (one delivery in transit per order). An instruction is refused if the slots it needs would take the total past 3,276 (80%), so every payout and delivery has its slot before it exists. A late top-up to a closed account waits in a restricted balance, owned by the client, until a slot is free, so slots in use never exceed 4,096. Synthetic capacity tests (Appendix E6; not compliant network scenarios) confirm both rules.

Table 9: Risks found during testing, ranked by harm and likelihood.

Risk	Evidence	Treatment
1 A solar conjunction cuts a client off from its host	During the 2028 Mars conjunction the Earth–Mars direct path is closed for 53 days. Earth clients cannot order, cancel or withdraw at Mars. Bob’s 30-day sell order expires first, so the trade does not happen (E5).	Rule R9 forwards instructions through a third exchange (Mercury here): the trade completes in 4.55 h and both futures runs complete. Without R9, the access check keeps money at home and nobody trades.
2 Money left at a host can become unreachable	Withdrawals are client instructions sent by direct service. Money left at Mars during a closure cannot be called home, even though the backbone still works.	Clients fund just before they trade (R7). Home delivery and payouts need no instruction. A withdrawal can be forwarded through a third exchange (R9).
3 Direct loss of 76–91% per copy at the outer settlements	From Neptune, $P(\text{complete within 72 h}) = 0.47$, because 8 copies only succeed 54% of the time (S3).	Change I1 (applied): up to 8 copies instead of 6, fallback after 6 h instead of 24 h, and resending until the order expires. This raised $P(72 \text{ h})$ from 0.34 to 0.47 for Neptune and from 0.93 to 0.99 for Uranus.
4 Concentration at the host	All futures margin and the price source are at Mars. An isolation of Mars around the fixing delays both payouts by up to 75 h (S2 search).	Fixing and payout are local and automatic. Payouts wait, already funded, in the outbox.
5 The access check ignored closures later in the sequence	With a 48 h check window, Mercury’s funding was stuck for 335 h because the Mercury–Mars direct path closes from h 22 to h 357 (S3).	Change I2 (applied): the check covers the actual length of the sequence. Mercury now completes in 3.81 h.
6 Sessions are silently lost after an exchange reset	The worst of 56,280 simulated incidents (S2) is a Mars reset at h 0.75, just after the handshakes. Mars silently drops the funding transfers, and the senders wait about 100 h for endpoint retries plus a 24 h backoff. The futures complete at 453.79 h instead of 328.17 h. The contract still opens before the deadline, and no money is at risk.	Alternative A1: after a reset, the exchange sends a restart notice to its recent peers. The futures then complete at 329.49 h. A1 is not part of the evaluated design; it is the first change for the next revision.
7 Status reports at scale	If Mars reported to each remote client by direct service, its 12 direct packets per day would cover at most 12 status changes a day for all clients together.	Status travels as batched STATUS records to home exchanges over the backbone (R6), at most one entry per client per 6 h unless the state changes. Clients still rely on fallbacks.
8 Single price source	Bob may fail or publish conflicting copies.	The fixing falls back to the latest valid earlier observation, or 100. This basis risk is disclosed.
9 Trust in the custodian exchange	Prefunding concentrates exposure on the host exchange.	The brief assumes operators follow the rules. Independent audit is future work.

Compaction without losing duplicate protection. A source numbers its transfers to each destination consecutively and sends them in that order. When a transfer is credited, the destination replaces the full record by a *tombstone*: the transfer identifier, the final state CREDITED and a 32-byte hash of the terms. A later copy that matches a tombstone only causes another RECEIVED, and a copy whose terms do not match the hash is rejected. For each source the destination also keeps a *watermark* W , the highest number such that every transfer up to W has been credited. Tombstones at or below W are dropped, because any

identifier at or below W is known to be a duplicate. A source does not send transfer n while any transfer numbered $n - 512$ or lower is unresolved, so every credited transfer above W lies within 512 of it: at most 512 tombstones per source, however long a gap stays open. The source compacts its own export to a tombstone when the RECEIVED arrives. Order and instruction identifiers are compacted the same way, and any copy that arrives after the order’s expiry or the contract’s opening deadline is rejected on that ground alone. The result of a withdrawal, funding, cancel or close request is kept for 60 days, twice the 30-day packet lifetime, and a copy older than 30 days is rejected. A 52-byte closure record diverts late top-ups until the host’s watermark passes the home’s last funding sent before it learned of the closure; a contract tombstone lasts until its instructions expire (Appendix E6; design rules, not simulated). Full records can move to an off-line audit archive; the live ledger needs only balances, open records, tombstones above the watermarks and one watermark per peer. When a limit is reached the exchange refuses new work; an obligation it has accepted already holds the capacity it needs.

Accepted demand. Each exchange can make up to 44 routine backbone originations per day, which is enough for about 10 new funding or withdrawal transfers between exchanges at 4 to 5 originations each, plus 22 originations for recovery. Each client has 12 direct packets per day, which is enough for 1 to 4 remote instructions depending on distance. A forwarding exchange (R9) also has 12 direct packets per day, so it can forward 1 to 4 instructions a day. Work beyond this is queued, and we make no promise about its timing.

Very long horizons. The connectivity argument and the delay bound above depend only on the shapes of the orbits, so they hold at every time in the model. The scanned statistics (availability, closure counts and typical delays) describe only the 200 scanned years. The orbital periods are not commensurate, so the joint configuration of the bodies never repeats exactly, and we do not extend those numbers beyond the scan. Our evidence for later epochs is the runs at +1, +10 and +100 years (Appendix E5). With integer milliseconds and extended-precision phase, nothing in the rules depends on the date. The remaining limits are the identifier space described above and the physical infrastructure, which the brief fixes.

7 Deployment assessment

The baseline model leaves out several real effects on purpose. None of them can break conservation or cause an asset to be used twice, because they only delay or lose packets, which the rules already tolerate. They would change our timings and probabilities. Table 10 estimates their size.

Table 10: Effects a real deployment would add.

Effect	Estimated size	Source and assumptions	Effect on the guarantees
Earth rotation	About 11.97 h without view per ground station per day	Sidereal day of 23.93 h, one equatorial station, no elevation mask [1]	Delays only, up to 12 h per hop with one station; G1–G5 unchanged.
Doppler shift	About 3.20 MHz at 32 GHz	$f \cdot v/c$ with an assumed line-of-sight speed of 30 km/s [1, 2]	Modems track it; otherwise more loss, which only slows service.
Relativistic clock rates	Mercury vs Earth: 2.3×10^{-8} , or 0.7 s per year	$GM/(rc^2) + v^2/(2c^2)$ at 0.39 AU and 1 AU	Maturity and fixing use TDB; local clocks are corrected.
Ephemeris drift	Not estimated	Kepler elements ignore perturbations between planets	Timetables need a live ephemeris.

References

- [1] NASA, Earth Fact Sheet. <https://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html>
- [2] ESA, Cebreros DSA-2 ground station (Ka-band, 32 GHz).
- [3] CPMI-IOSCO, *Principles for Financial Market Infrastructures* (delivery versus payment), 2012.
- [4] J. Gray and L. Lamport, Consensus on Transaction Commit, *ACM TODS* 31(1), 2006.